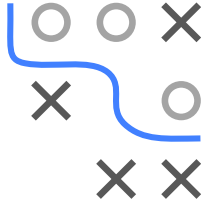
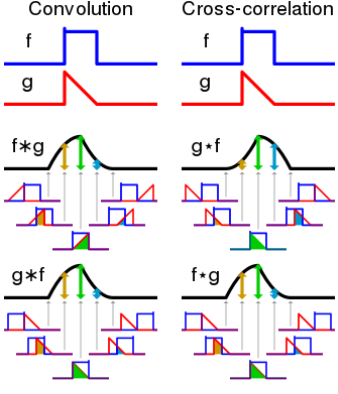
Introduction to Deep Learning



Convolutional Neural Networks I
Convolutions - Mathematical Perspective

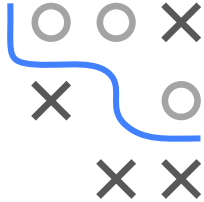


Learning goals

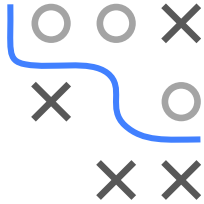
- Convolution vs. Cross-Correlation

CONVOLUTIONS : A DEEPER LOOK I

- CNNs borrow their name from a mathematical operation termed **convolution** that originates in Signal Processing.
- Basic understanding of this concept and related operations improves the understanding of the CNN functionality.



CONVOLUTIONS : A DEEPER LOOK II



- Definition:

$$h(i) = (f * g)(i) = \int f(x)g(i-x)dx$$

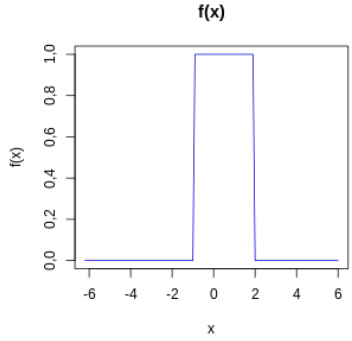
where $f(x)$: input function

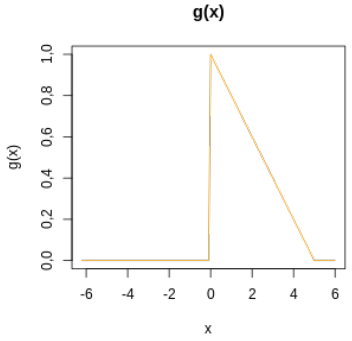
and $g(x)$: weighting function, kernel

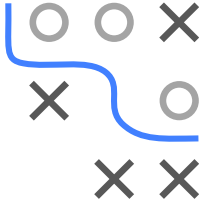
and $h(i)$: output function, feature map elements

- Intuition 1: weighted smoothing of $f(x)$ with weighting function $g(x)$.
- Intuition 2: filter function $g(x)$ filters features $h(i)$ from input signal $f(x)$.

1D CONVOLUTION ANIMATION I

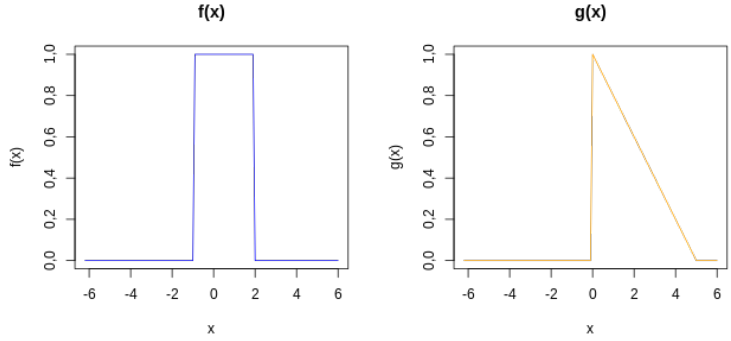


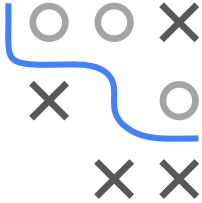




$$f(x) = \begin{cases} 1, & \text{if } x \in [-1, 2] \\ 0, & \text{otherwise} \end{cases} \qquad g(x) = \begin{cases} 1 - 0.2 * |x|, & \text{if } x > 0 \\ 0, & \text{otherwise} \end{cases}$$

1D CONVOLUTION ANIMATION I

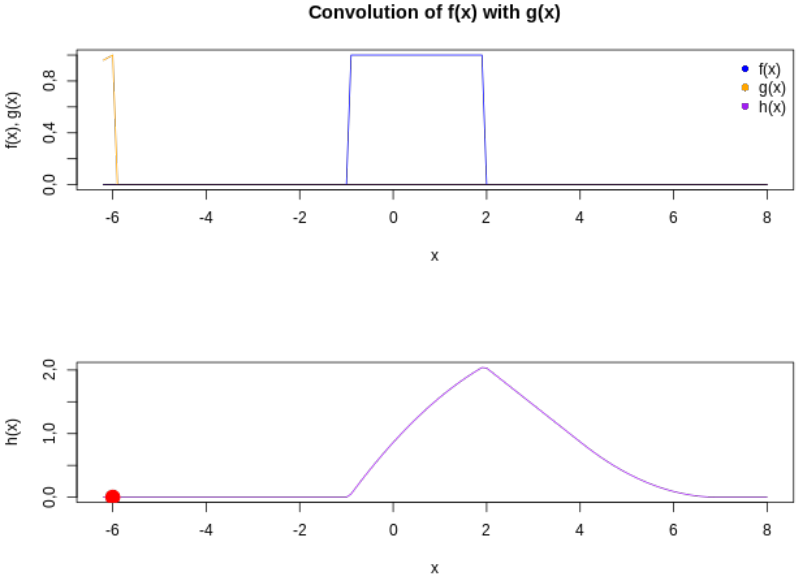


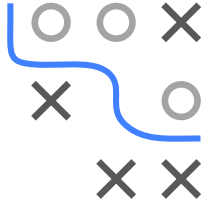


Kernel is flipped due to the negative iterator in

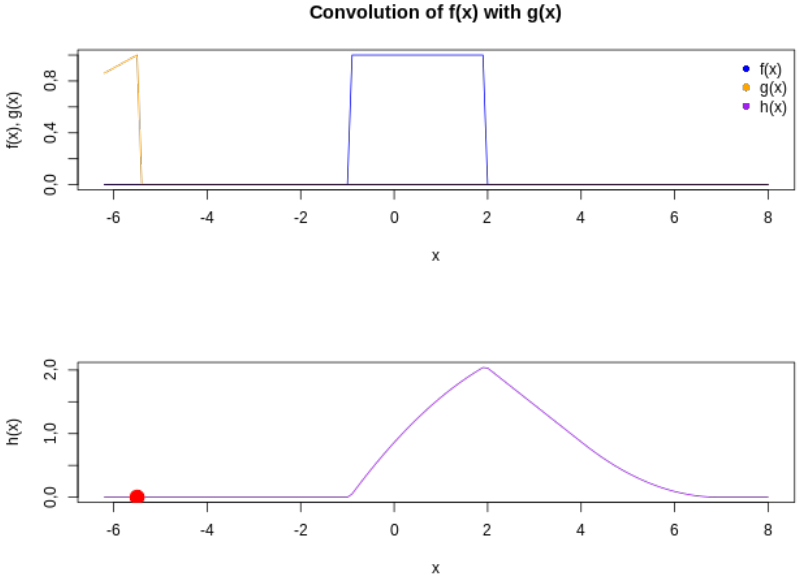
$$h(i) = \int_{x=-\infty}^{\infty} f(x)g(i-x)$$

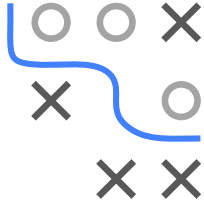
1D CONVOLUTION ANIMATION



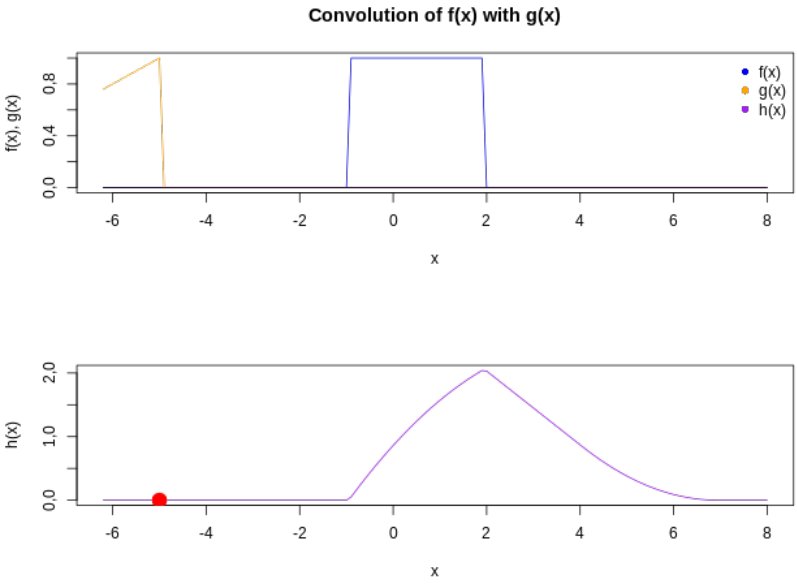


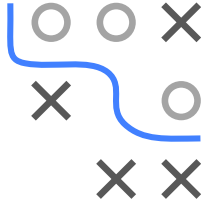
1D CONVOLUTION ANIMATION



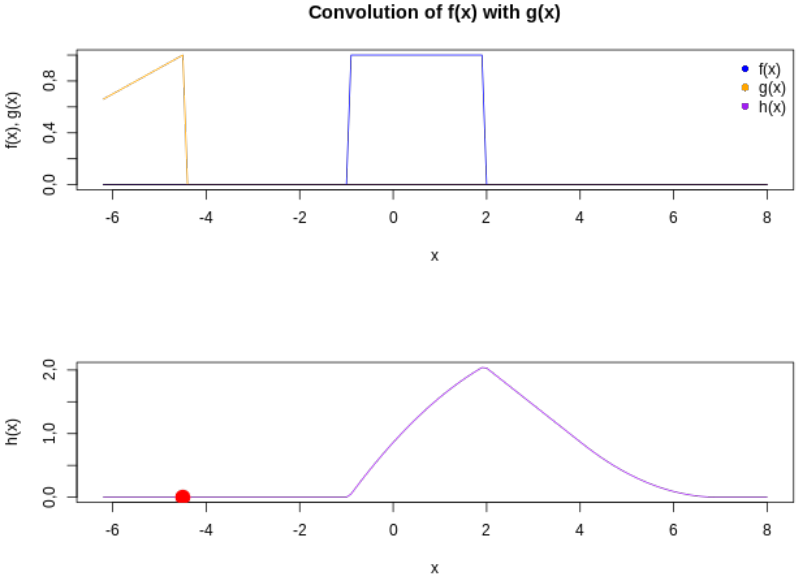


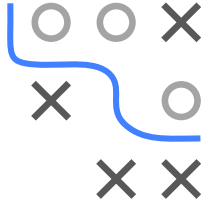
1D CONVOLUTION ANIMATION



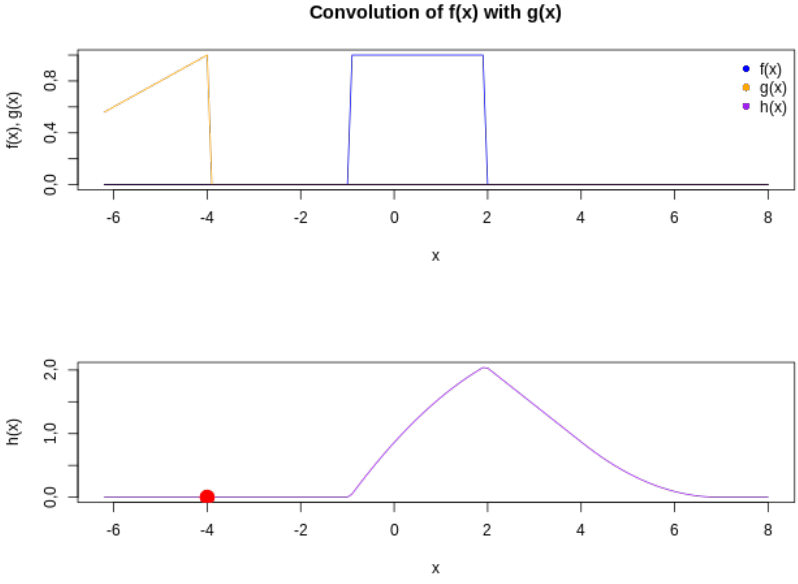


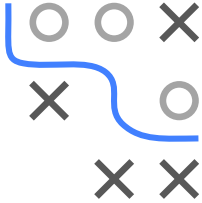
1D CONVOLUTION ANIMATION



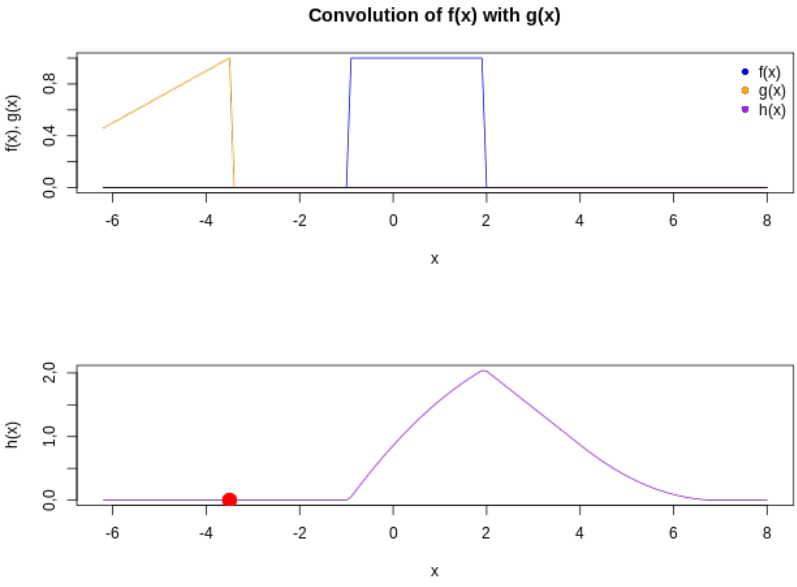


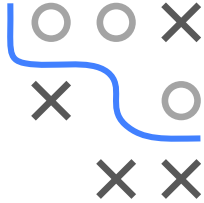
1D CONVOLUTION ANIMATION



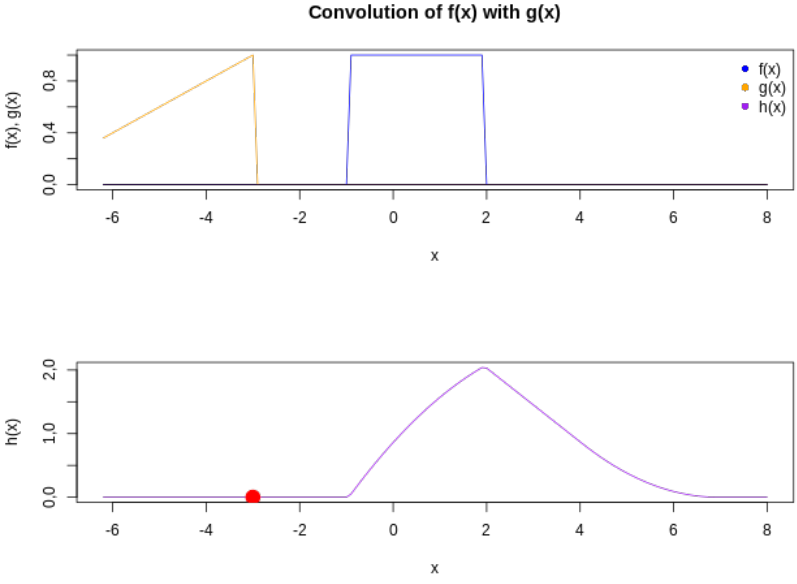


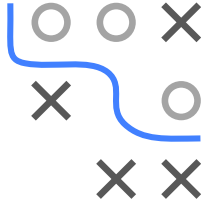
1D CONVOLUTION ANIMATION



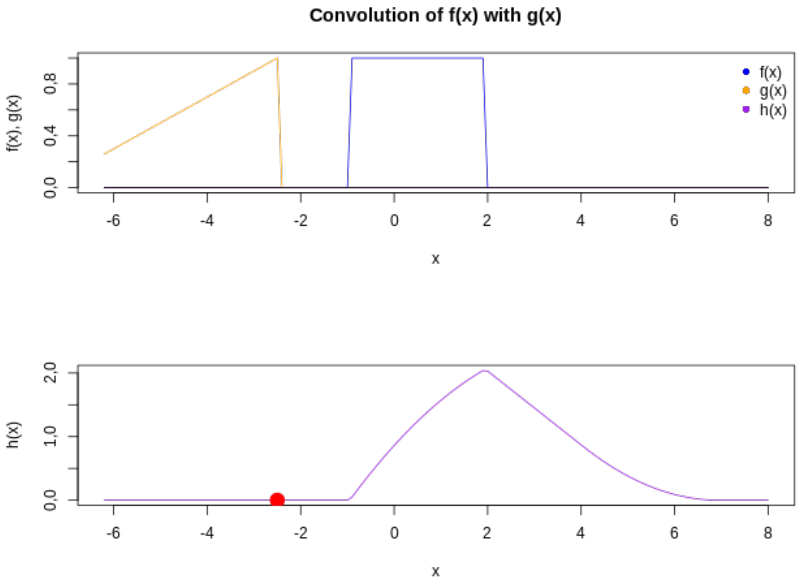


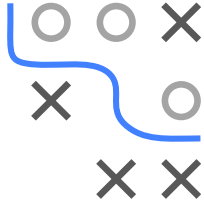
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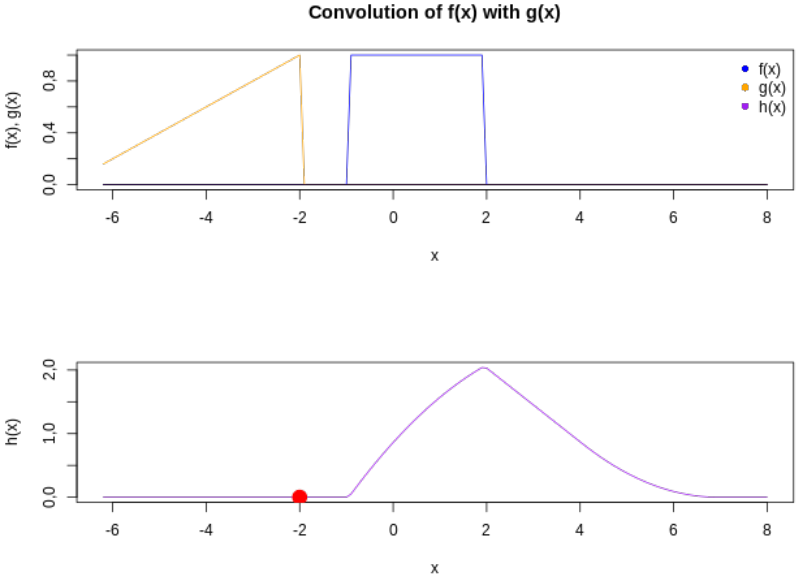


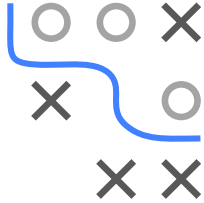
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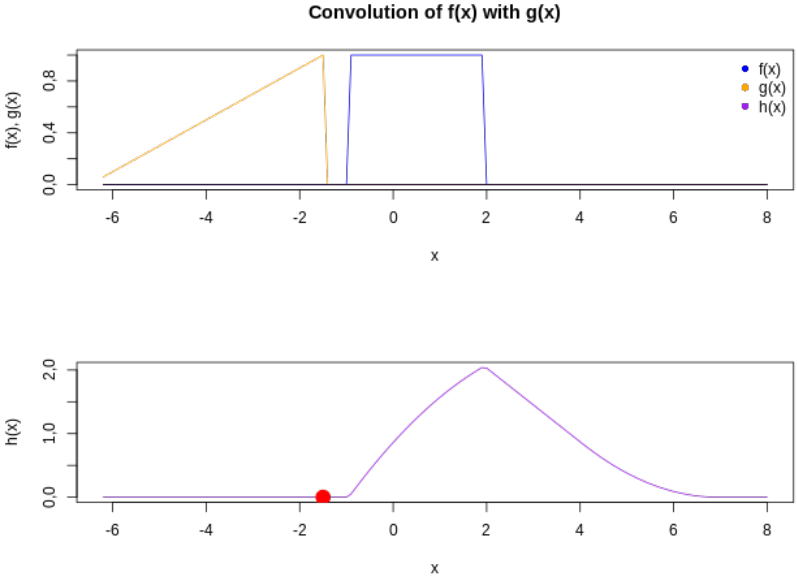


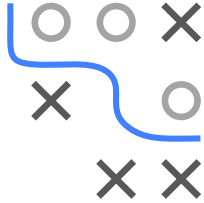
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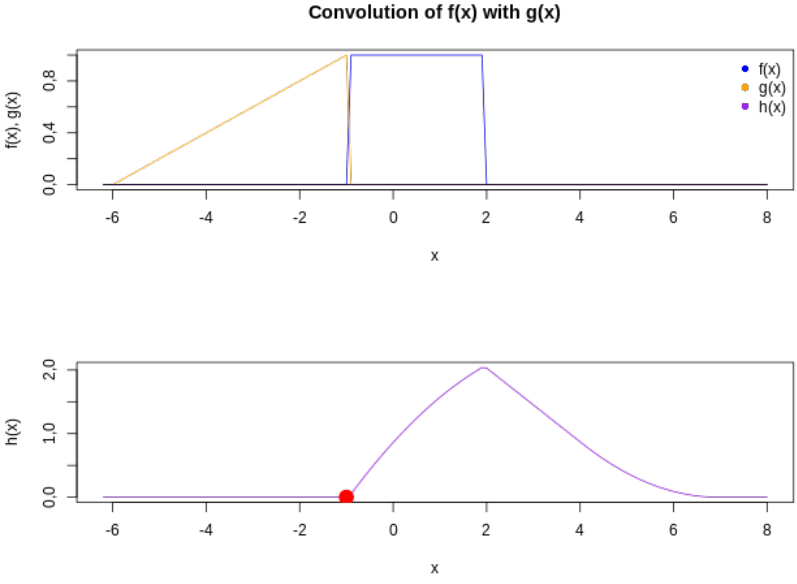


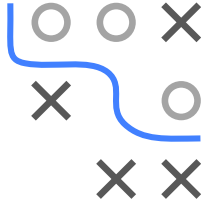
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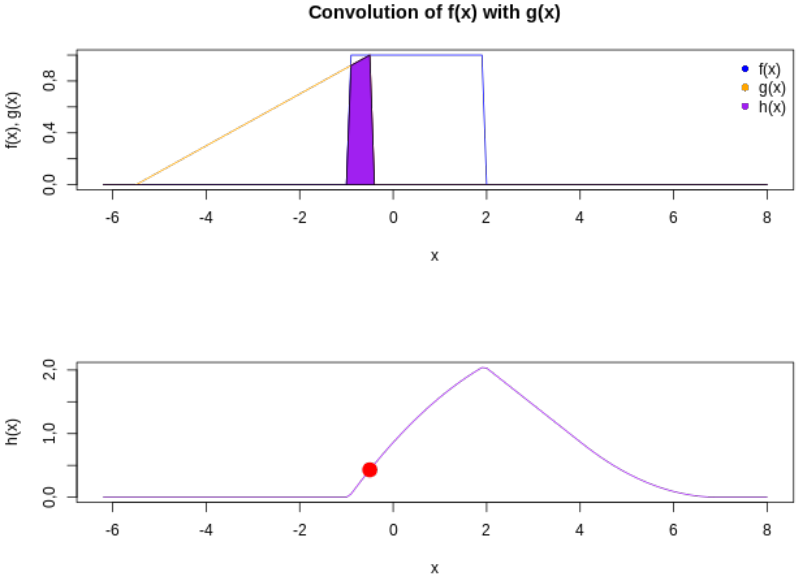


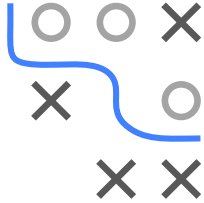
1D CONVOLUTION ANIMATION



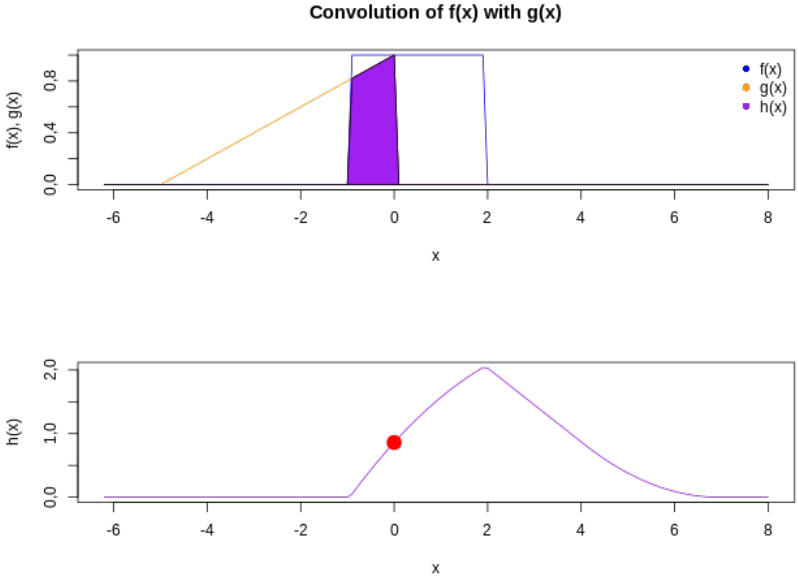


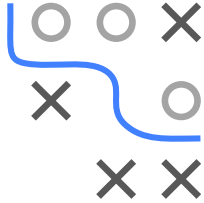
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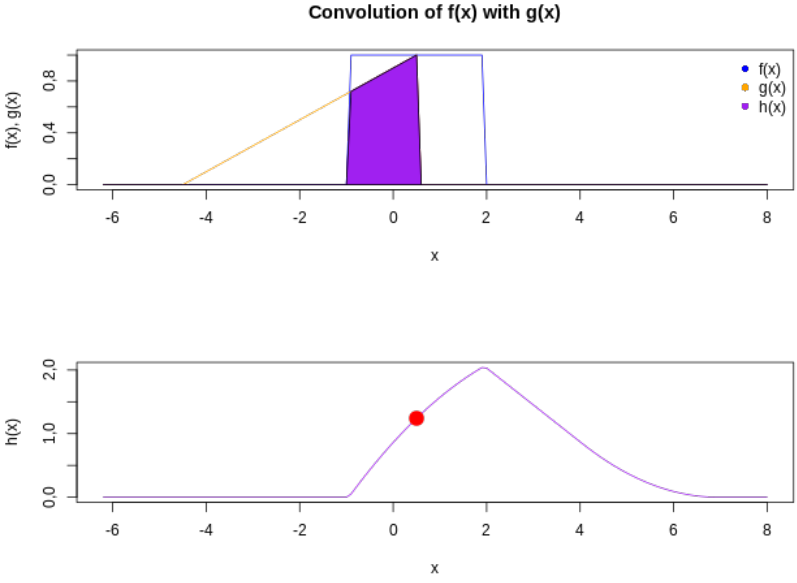


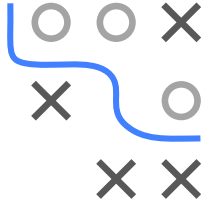
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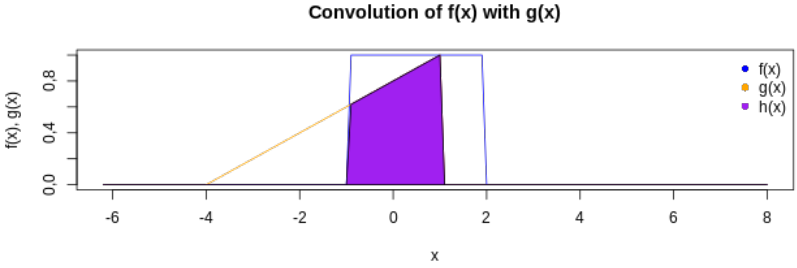


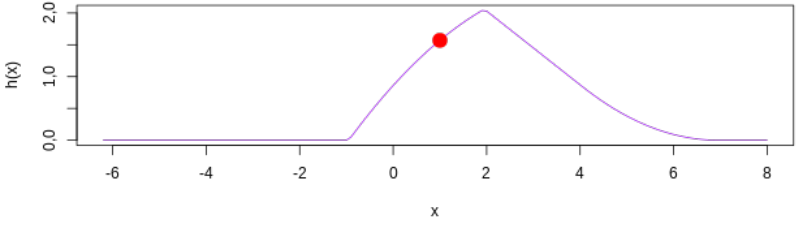
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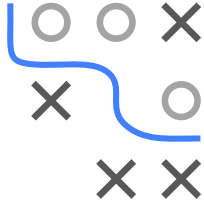




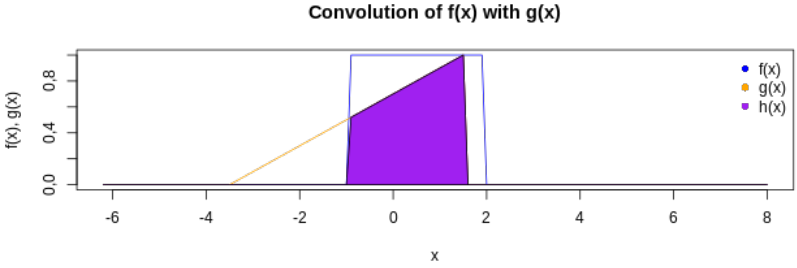
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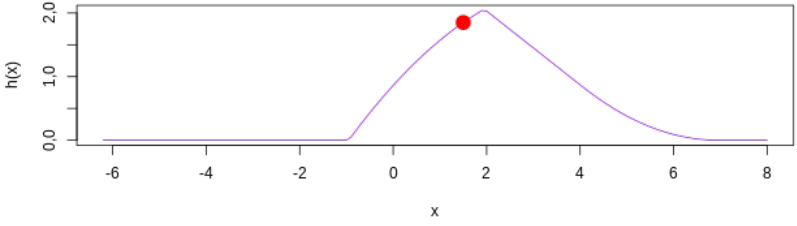


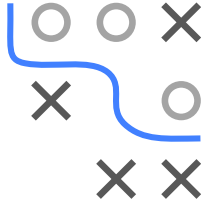




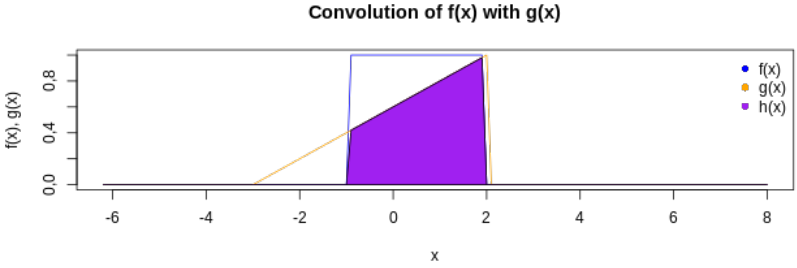
1D CONVOLUTION ANIMATION

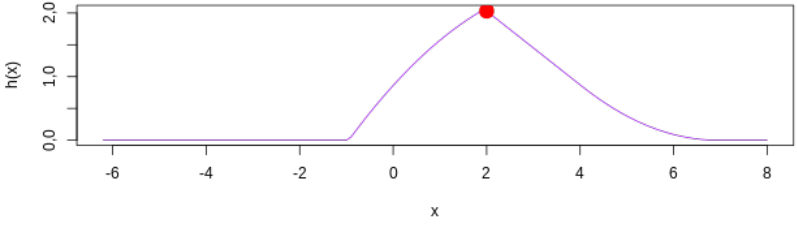


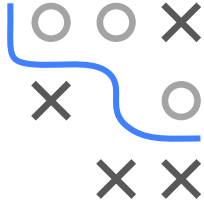




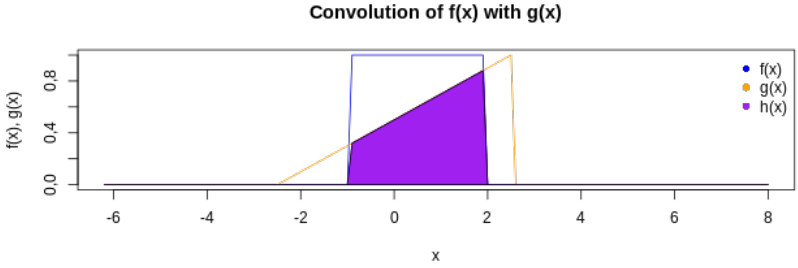
1D CONVOLUTION ANIMATION

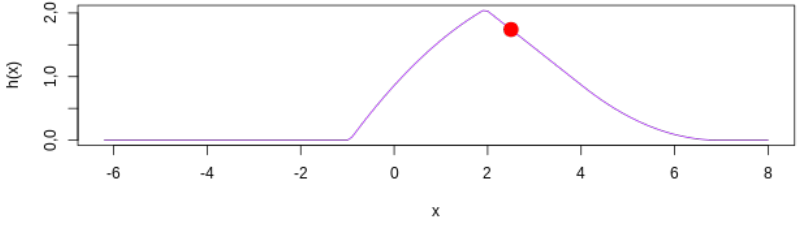


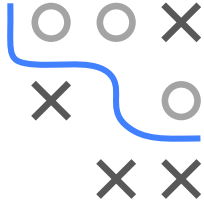




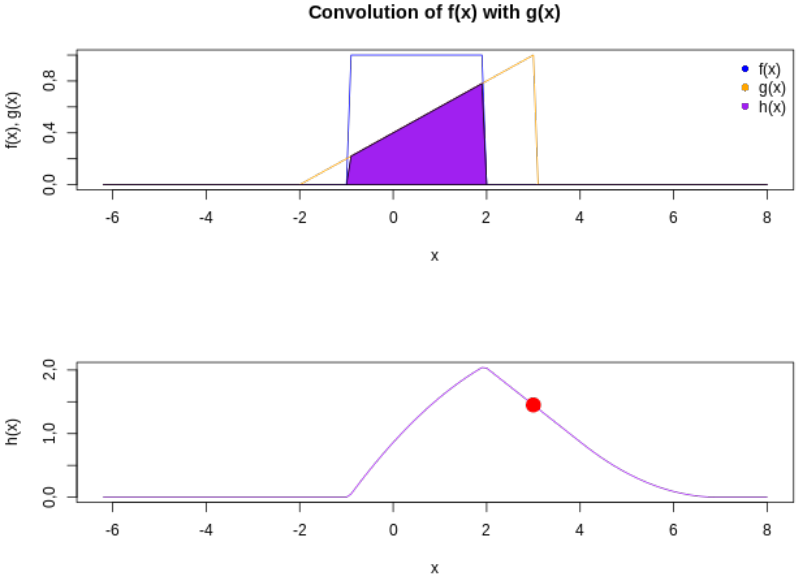
1D CONVOLUTION ANIMATION

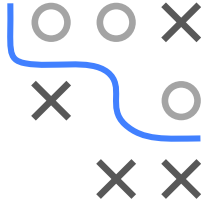




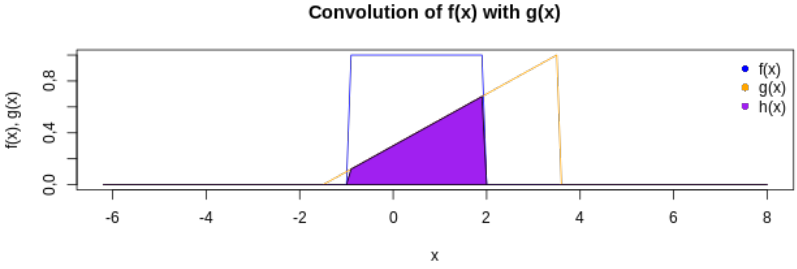


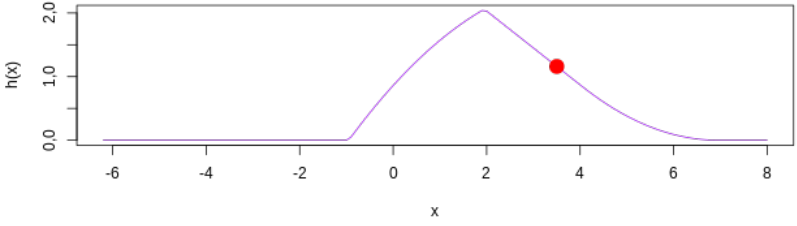
1D CONVOLUTION ANIMATION

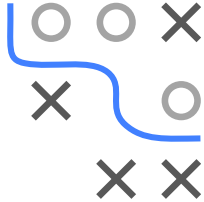




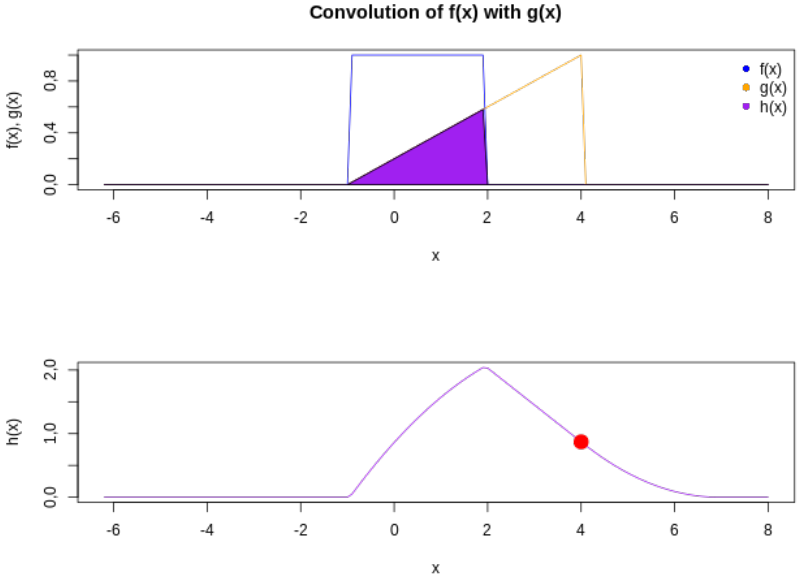
1D CONVOLUTION ANIMATION

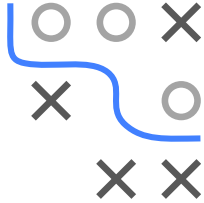




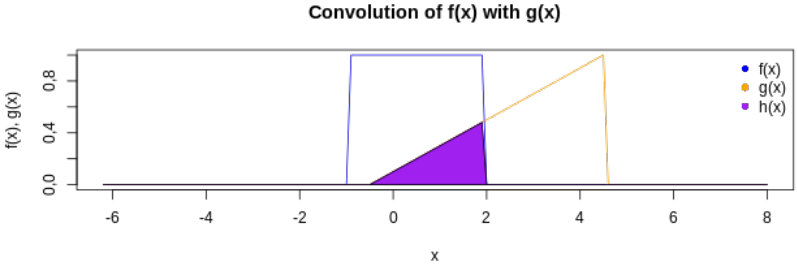


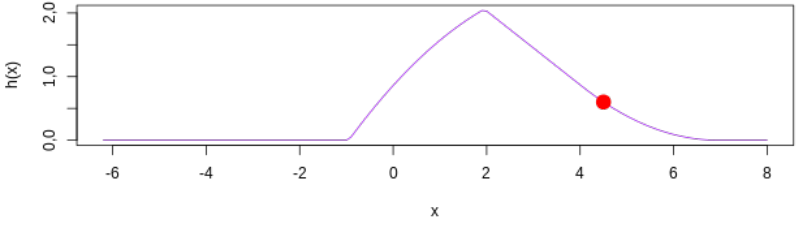
1D CONVOLUTION ANIMATION

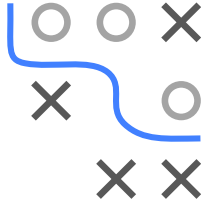




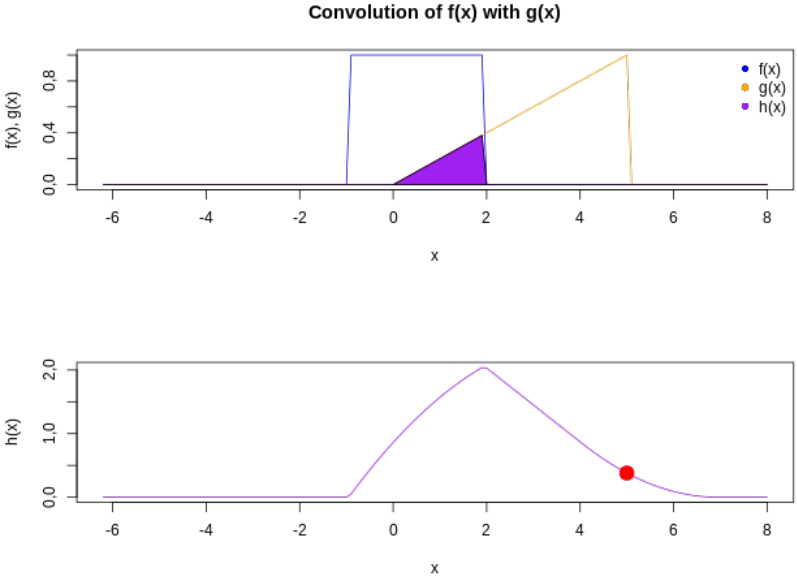
1D CONVOLUTION ANIMATION

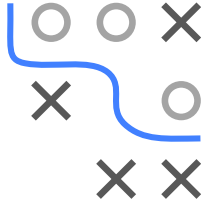




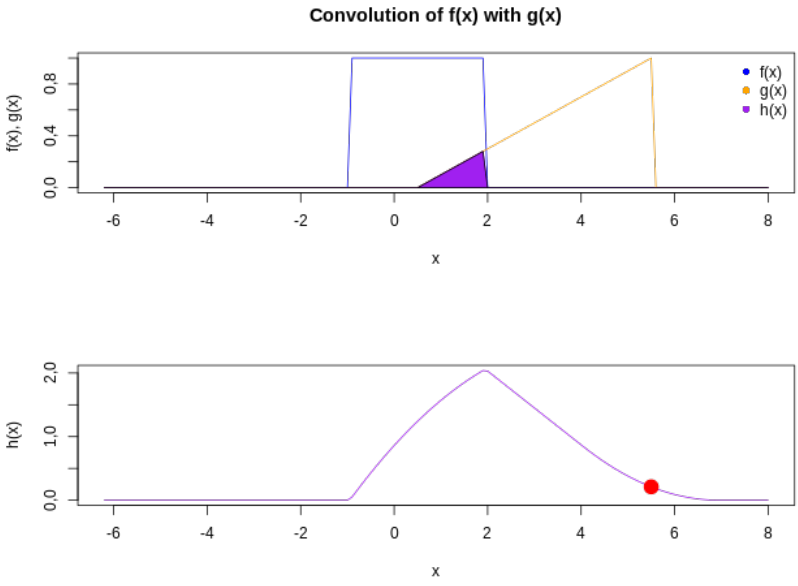


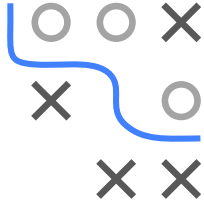
1D CONVOLUTION ANIMATION



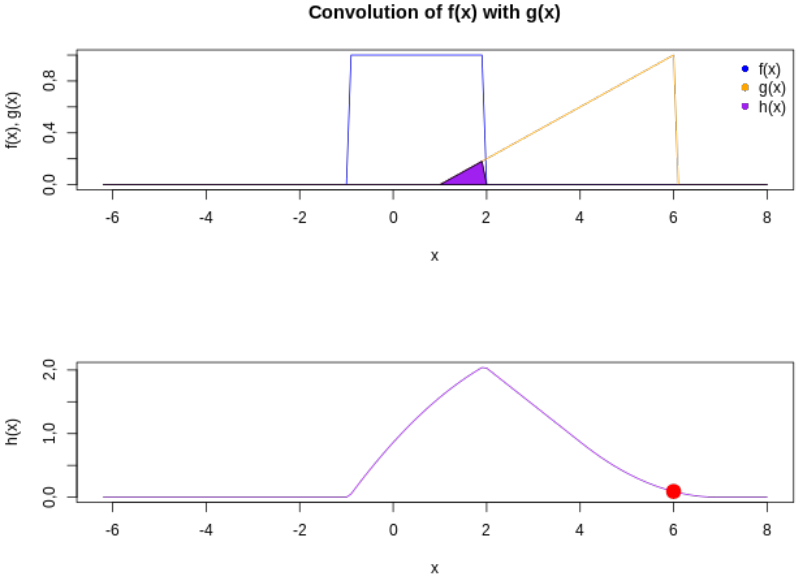


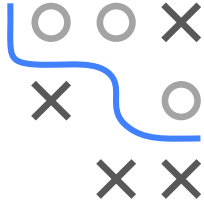
1D CONVOLUTION ANIMATION



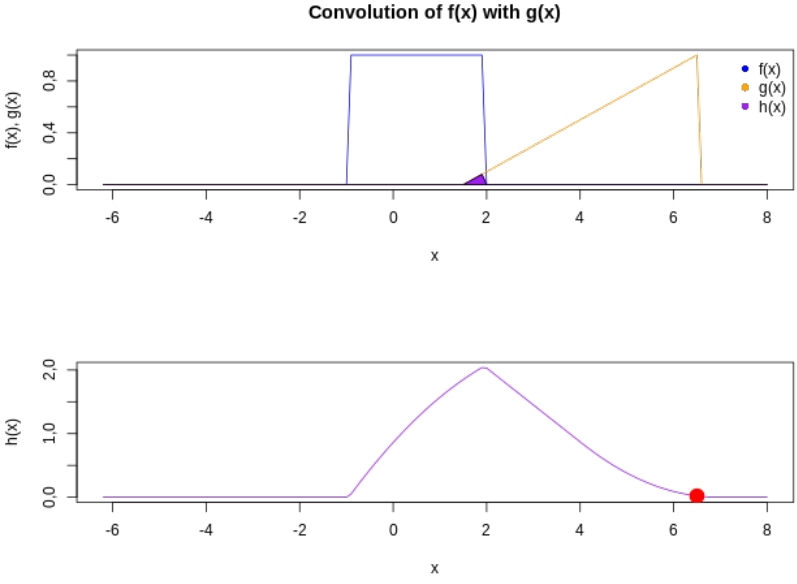


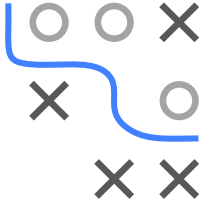
1D CONVOLUTION ANIMATION



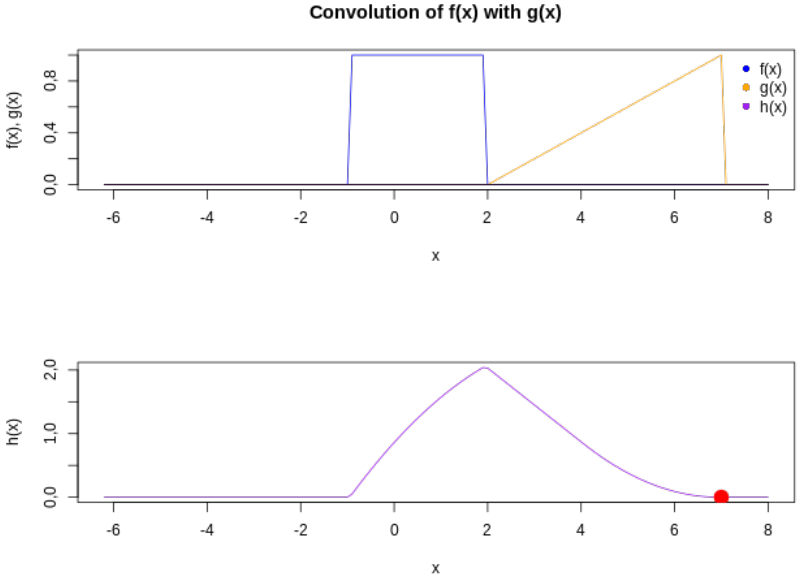


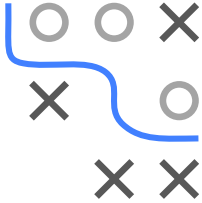
1D CONVOLUTION ANIMATION



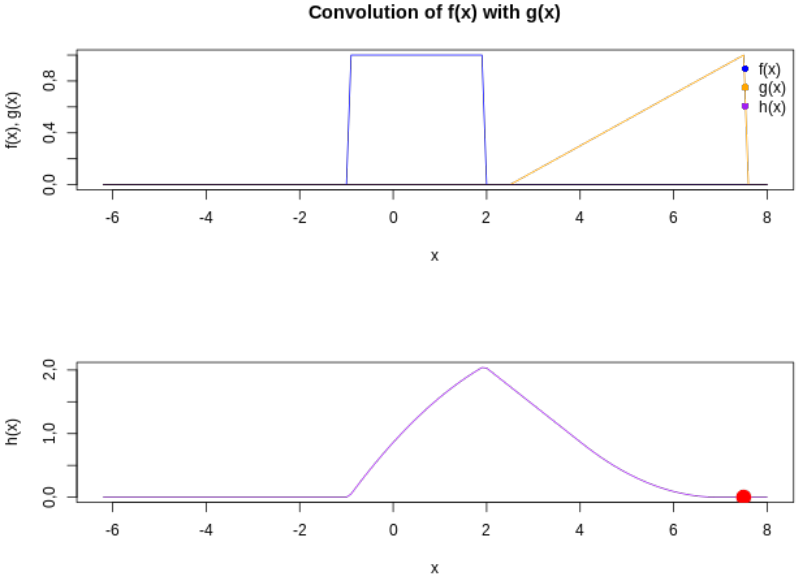


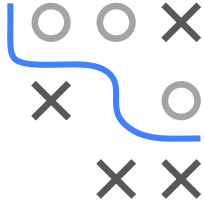
1D CONVOLUTION ANIMATION



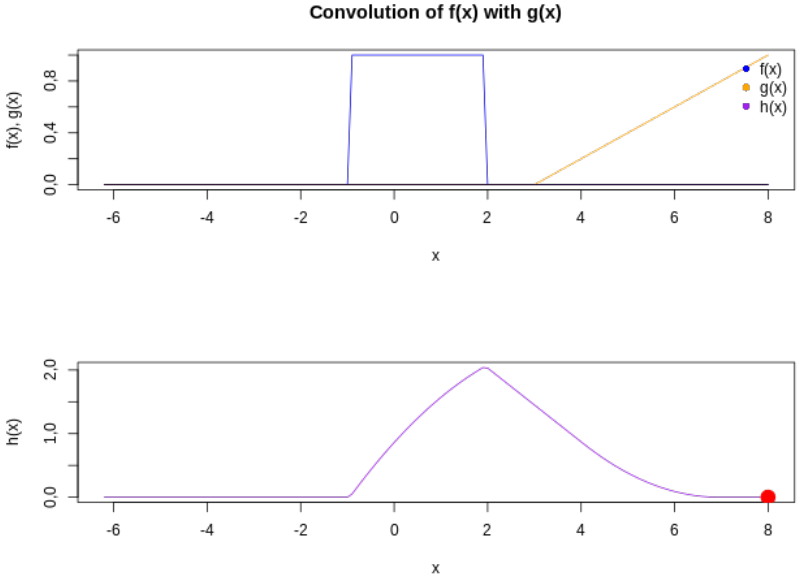


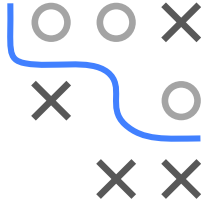
1D CONVOLUTION ANIMATION



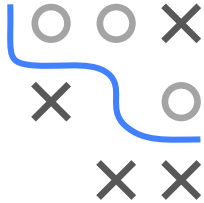


1D CONVOLUTION ANIMATION





DISCRETIZATION



- Discretization for one-dimensional input:

$$h(i) = (f * g)(i) = \sum_x f(x)g(i-x)$$

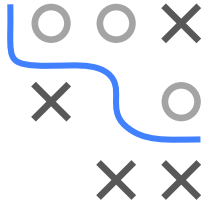
- Discretization for 2D images:
  - $\mathcal{I} \in \mathcal{R}^2$ contains two dimensions
  - Use 2D Kernel $\mathcal{G}$ as well to yield feature map $\mathcal{H}$:

$$H(i,j) = (\mathcal{I} * \mathcal{G})(i,j) = \sum_x \sum_y \mathcal{I}(x,y)\mathcal{G}(i-x, j-y)$$

where $x, y :=$ indices $\mathcal{I}$ and $\mathcal{G}$

and $i, j :=$ indices elements in $\mathcal{H}$

PROPERTIES OF THE CONVOLUTION I

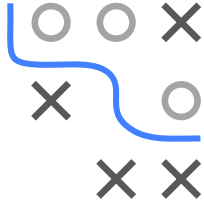


- Commutativity:
$$f * g = g * f$$
- Associativity:
$$(f * g) * h = f * (g * h)$$
- Distributivity:
$$f * (g + h) = f * g + f * h$$
$$\alpha(f * g) = (\alpha f) * g \text{ for scalar } \alpha$$
- Differentiability:
$$\frac{\partial (f * g)(x)}{\partial x_i} = \frac{\partial f(x)}{\partial x_i} * g(x) = \frac{\partial g(x)}{\partial x_i} * f(x)$$

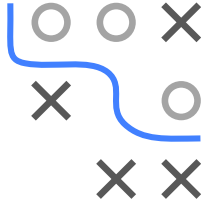
  $\rightarrow (f * g)(x)$ is as many times differentiable as the max of $g(x)$ and $f(x)$.

RELATED OPERATIONS I

- Convolution is strongly related to two other mathematical operators:
  1. Fourier transform via the Convolution Theorem
  2. Cross correlation



CONVOLUTION THEOREM I



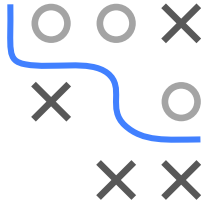
- Fourier transform of the convolution of two functions can be expressed as the product of their Fourier transforms:

$$\mathcal{F}\{f * g\} = \mathcal{F}\{f\}\mathcal{F}\{g\}$$

- Transformation of a signal from time to frequency domain.
- Convolution in the time domain is equivalent to multiplication in frequency domain.
- The computationally fastest way to compute a convolution is therefore to take the Fourier inverse of the multiplication of the Fourier-transformed input and filter function :

$$(f * g)(t) = \mathcal{F}^{-1}\{\mathcal{F}\{f(t)\}\mathcal{F}\{g(t)\}\}$$

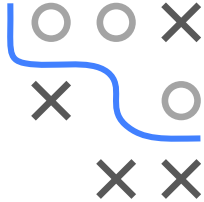
CONVOLUTION THEOREM - PROOF I



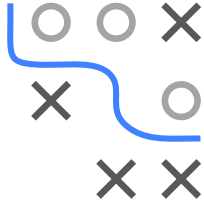
$$
\begin{aligned}
\widehat{(f * g)}(t) &= \int_{-\infty}^{\infty} exp(-2\pi i\omega t)\Big[\int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau\Big] \\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} exp(-2\pi i\omega t)f(\tau)g(t-\tau)d\tau dt \\
&\overset{Fubini}{=} \int_{-\infty}^{\infty}\Big[\int_{-\infty}^{\infty} exp(-2\pi i\omega t)f(\tau)g(t-\tau)dt\Big]d\tau \\
&\overset{f(\tau)\perp t}{=} \int_{-\infty}^{\infty} f(\tau)\Big[\int_{-\infty}^{\infty} exp(-2\pi i\omega t)g(t-\tau)dt\Big]d\tau \\
&\overset{u=t-\tau}{=} \int_{-\infty}^{\infty} f(\tau)\Big[\int_{-\infty}^{\infty} exp(-2\pi i\omega \tau)exp(-2\pi i\omega u)g(u)du\Big]d\tau \\
&= \int_{-\infty}^{\infty} exp(-2\pi i\omega \tau)f(\tau)\Big[\int_{-\infty}^{\infty} exp(-2\pi i\omega u)g(u)du\Big]d\tau \\
&\overset{Fubini}{=} \dots
\end{aligned}
$$

CONVOLUTION THEOREM - PROOF I

$$\ldots \int_{-\infty}^{\infty} exp(-2\pi i\omega\tau) f(\tau) d\tau \int_{-\infty}^{\infty} exp(-2\pi i\omega u) g(u) du$$
$$= f(\hat{t}) g(\hat{t})$$



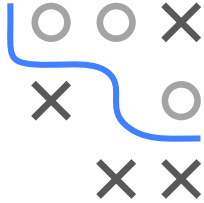
CROSS CORRELATION I



- Measurement for similarity of two functions $f(x), g(x)$.
- More specifically, at which position are the two functions most similar to each other? Where does the pattern of $g(x)$ match $f(x)$ the best?
- Intuition:
  - Slide with $g(x)$ over $f(x)$ and at each discrete step compute the sum of the product of their elements.
  - When peaks of both functions are aligned, the product of high (positive or negative) values will lead to high sums.
  - Thus, both functions are most similar at points with equal peaks.

CROSS CORRELATION II

- Definition:



$$h(i) = (f \star g)(i) = \int_{-\infty}^{\infty} f(x)g(i+x)dx$$

where $f(x)$: input function

and $g(x)$: weighting function, kernel

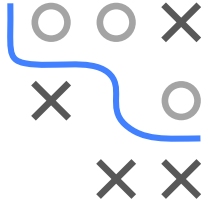
and $h(i)$: output function, feature map elements

for $f, g \in \mathcal{R}^d \mapsto h \in \mathcal{R}^d$

CROSS CORRELATION III

- Discrete formulation:

$$h(i) = (f \star g)(i) = \sum_{x=-\infty}^{\infty} f(x)g(i+x)$$

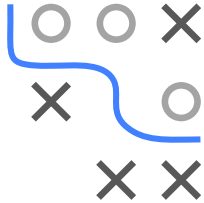


- Thus:

$$f(i) \star g(i) = f(-i) * g(i)$$

- Remember: $*$ is used for convolution and $\star$ for cross correlation.
- Similar formulation as the convolution despite the flipped filter function in the convolutional kernel .

CROSS CORRELATION IV



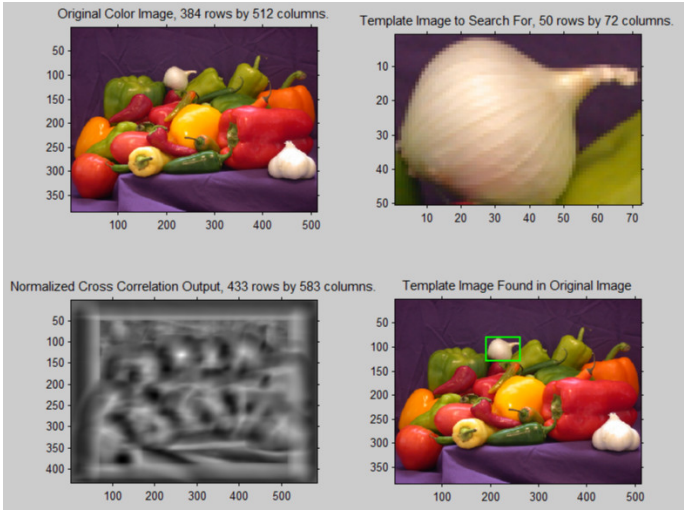
- This operation also works in 2 dimensions
- The difference w.r.t. the convolution are the positive iterators in the sum:

$$H(i,j) = (\mathcal{I} \star \mathcal{G})(i,j) = \sum_x \sum_y \mathcal{I}(x,y)\mathcal{G}(i+x, j+y)$$

where $x, y :=$ indices $\mathcal{I}$ and $\mathcal{G}$

and $i, j :=$ indices elements in $\mathcal{H}$

CROSS CORRELATION V



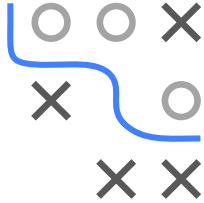
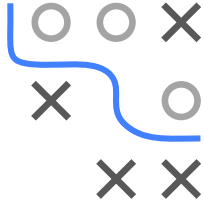


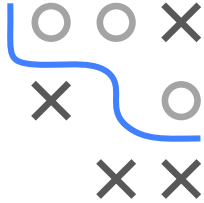
Figure: Cross-correlation used to detect a template (onion) in an image. Cross correlation peaks (white) at the position where template and input match best.

CROSS CORRELATION VI

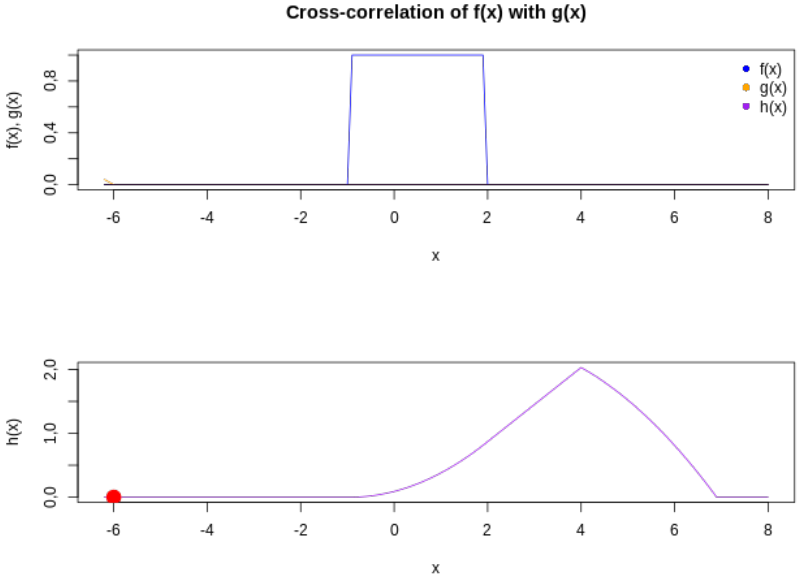


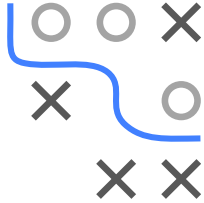
CROSS CORRELATION I

- From the following animation we see that
  - Kernel is not flipped as opposed to the convolution.
  - Cross-Correlation peaks, where the filter matches the signal the most.
- In some frameworks, Cross-Correlation is implemented instead of the convolution due to
  - better computational performance.
  - similar properties, as the kernel weights are learned throughout the training process.

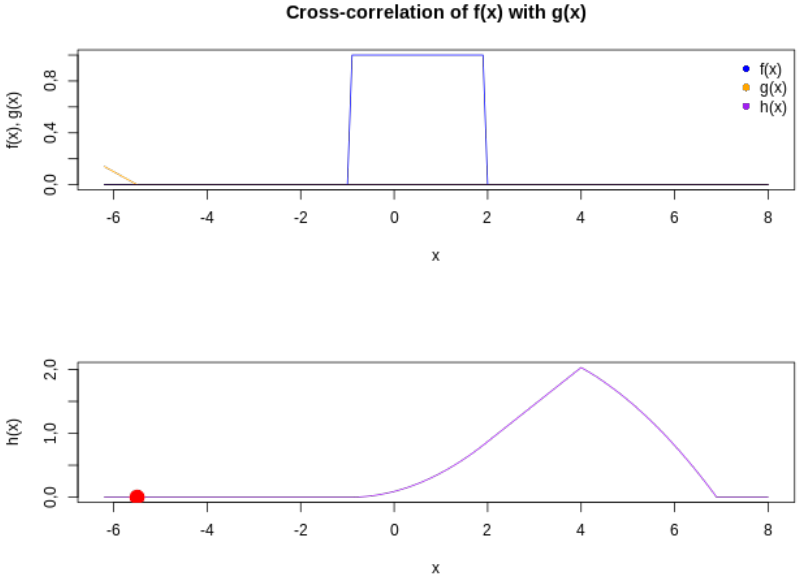


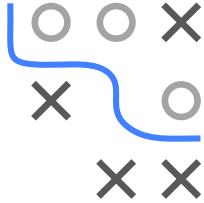
1D CROSS CORRELATION ANIMATION



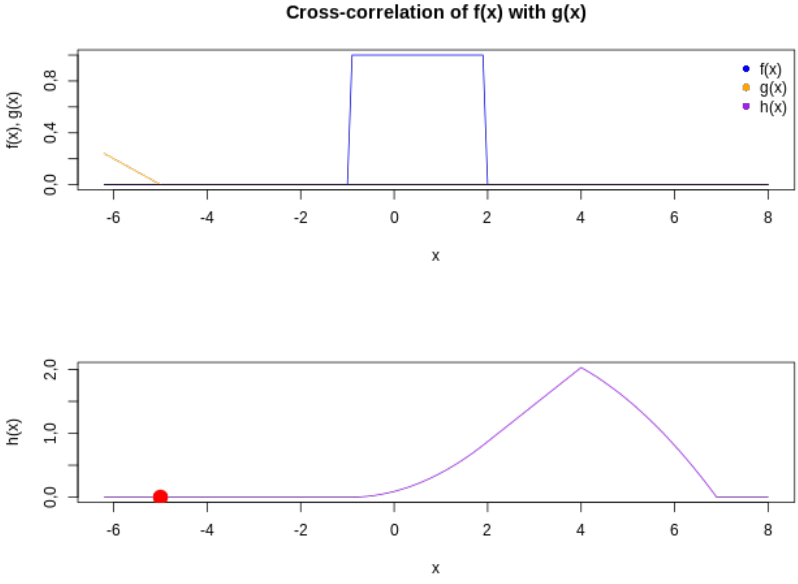


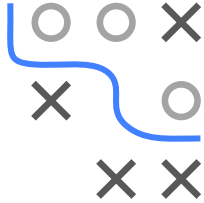
1D CROSS CORRELATION ANIMATION



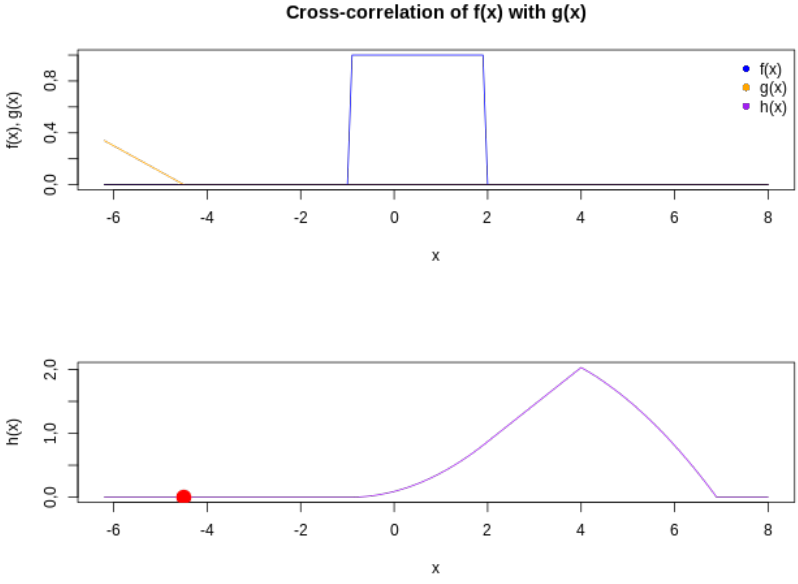


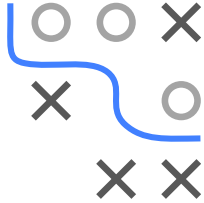
1D CROSS CORRELATION ANIMATION



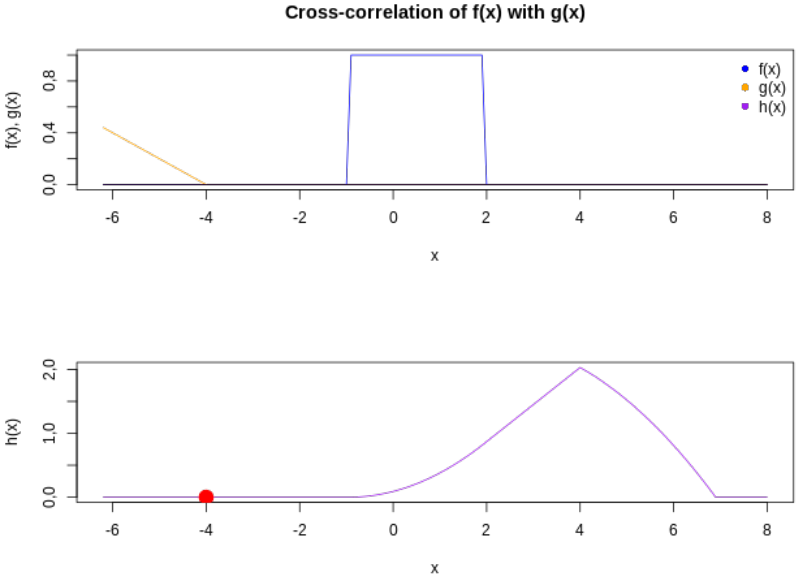


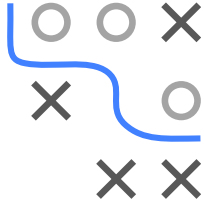
1D CROSS CORRELATION ANIMATION



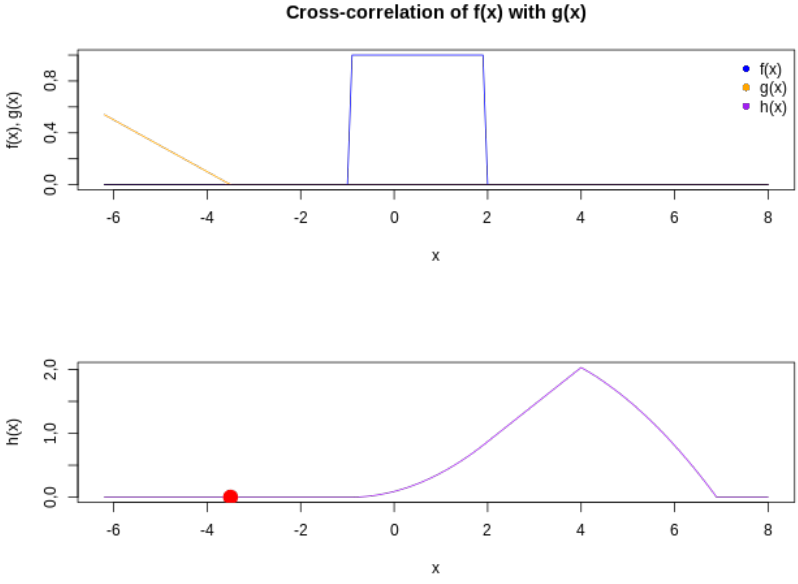


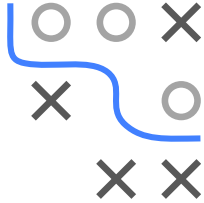
1D CROSS CORRELATION ANIMATION



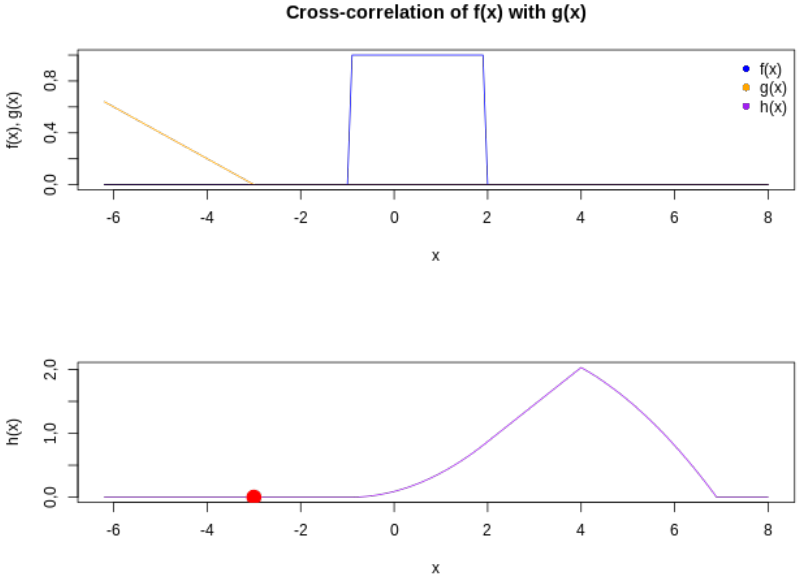


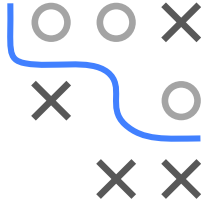
1D CROSS CORRELATION ANIMATION



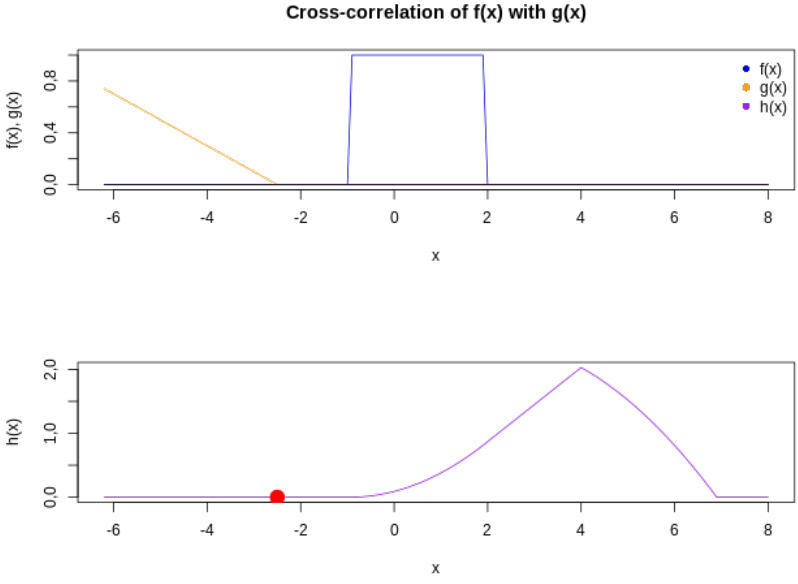


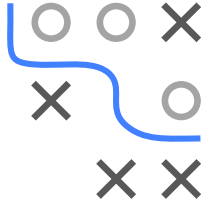
1D CROSS CORRELATION ANIMATION



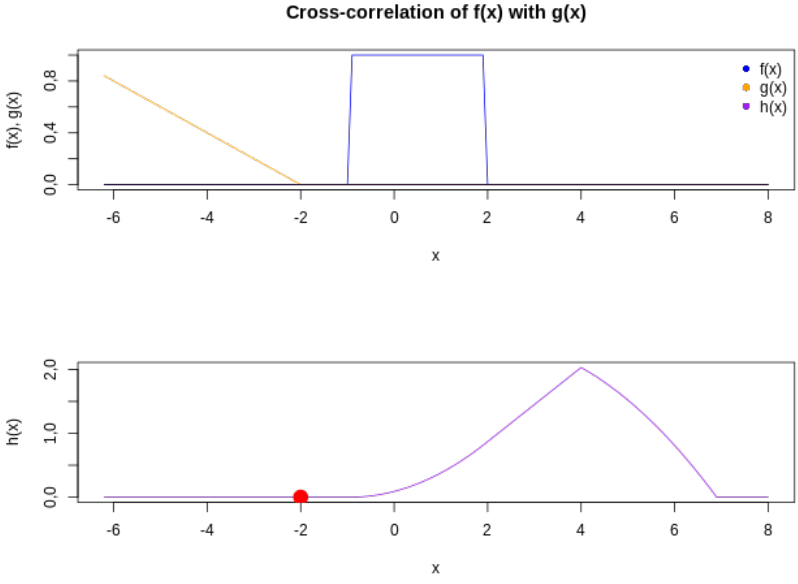


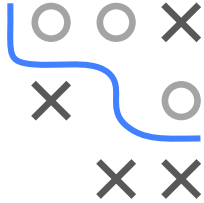
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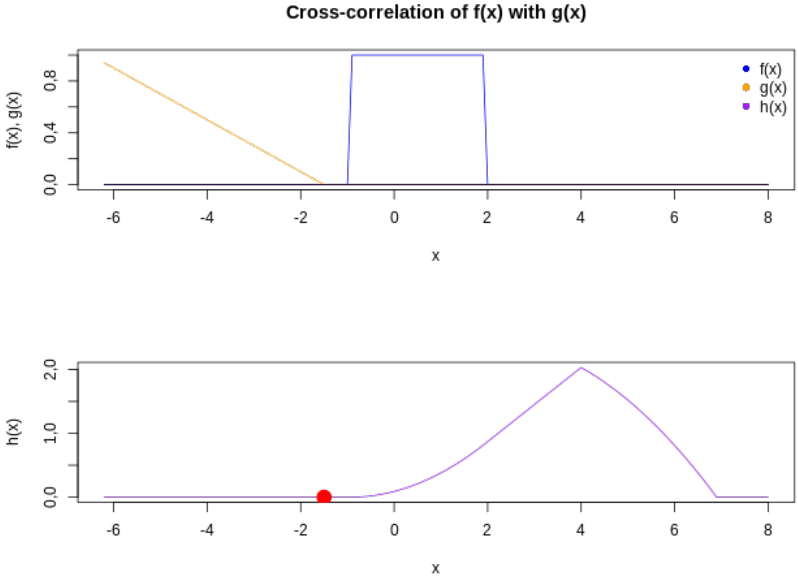


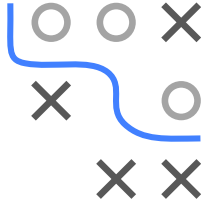
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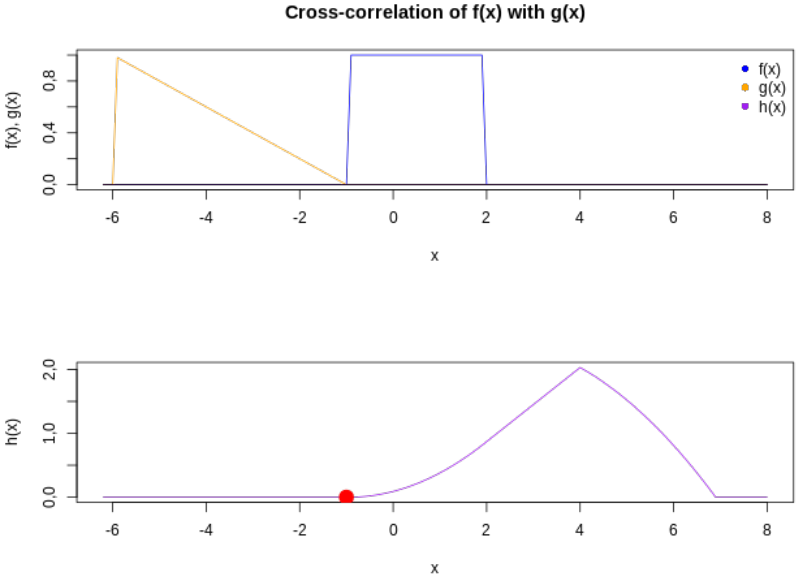


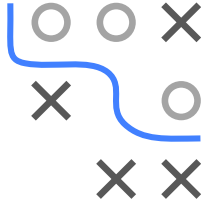
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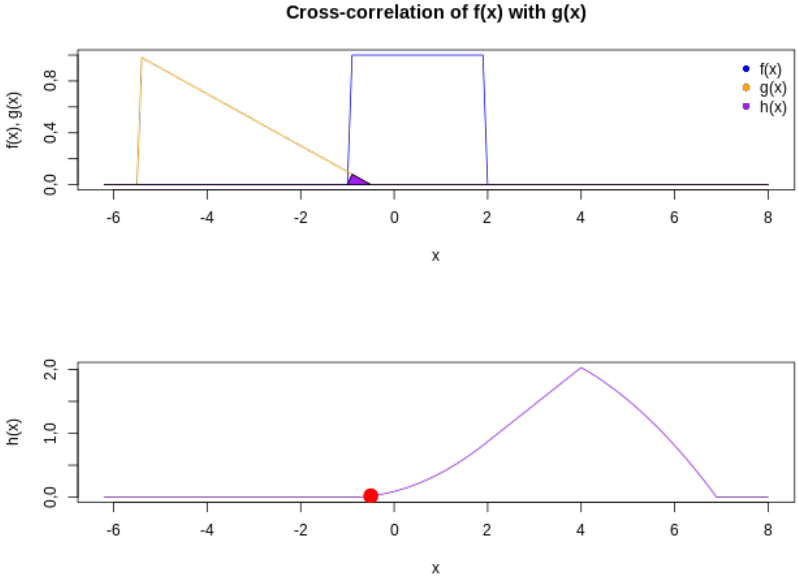


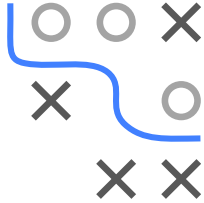
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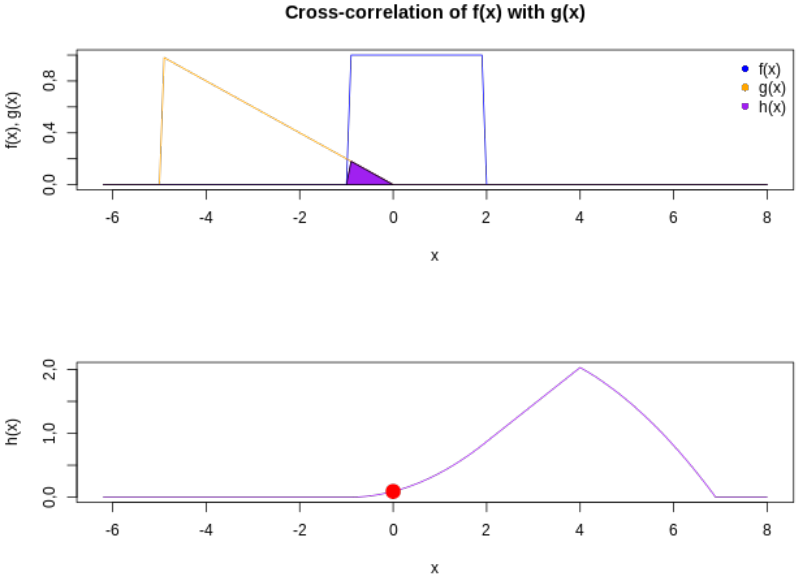


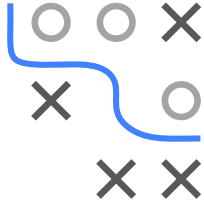
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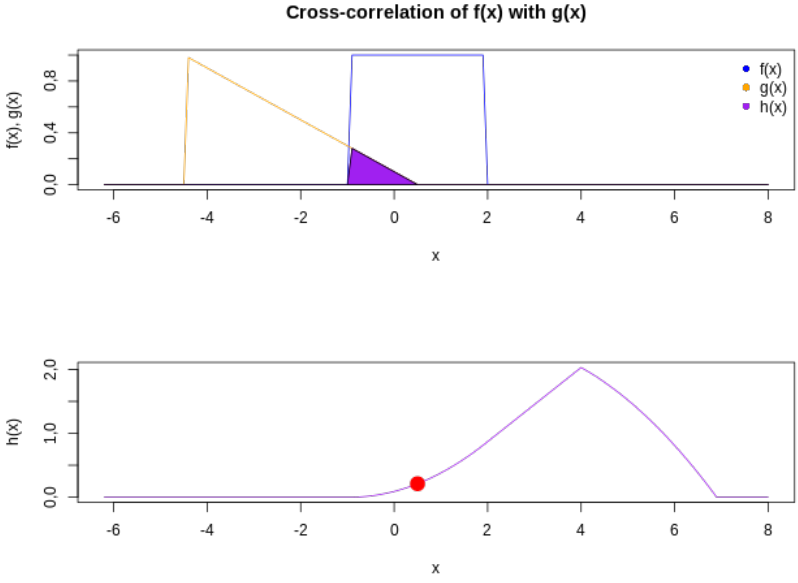


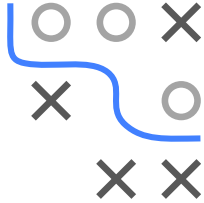
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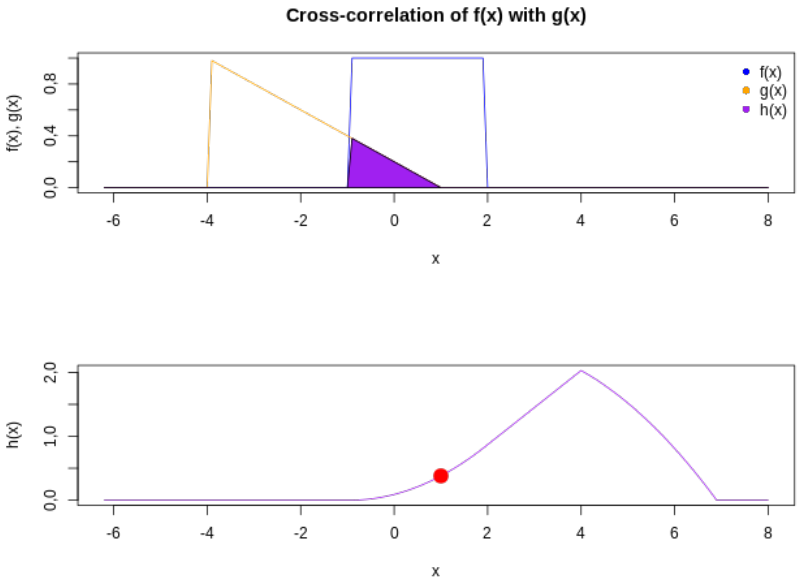


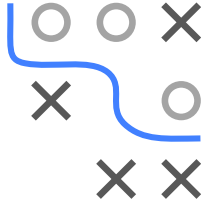
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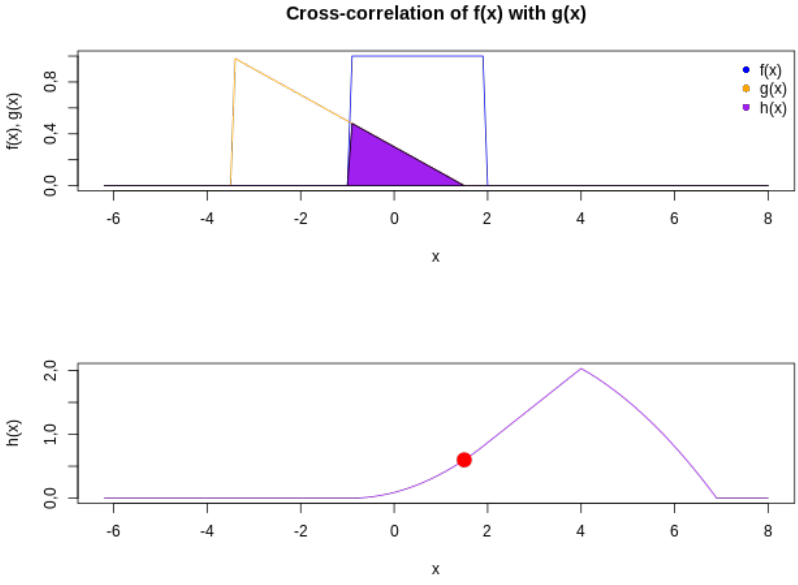


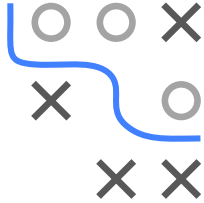
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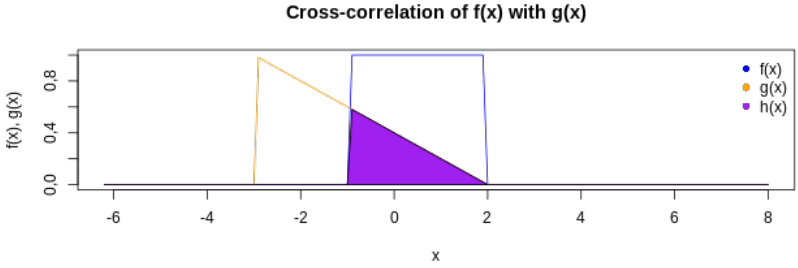


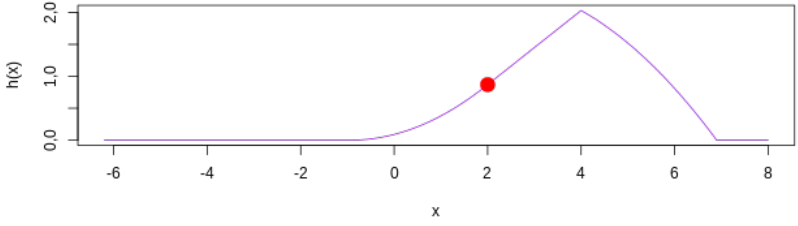
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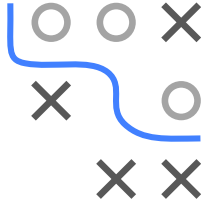




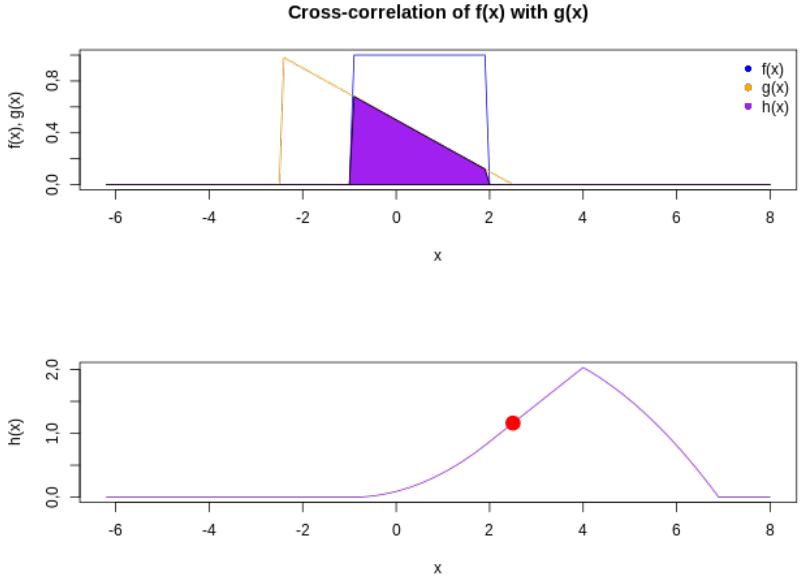
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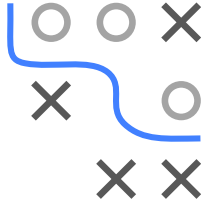




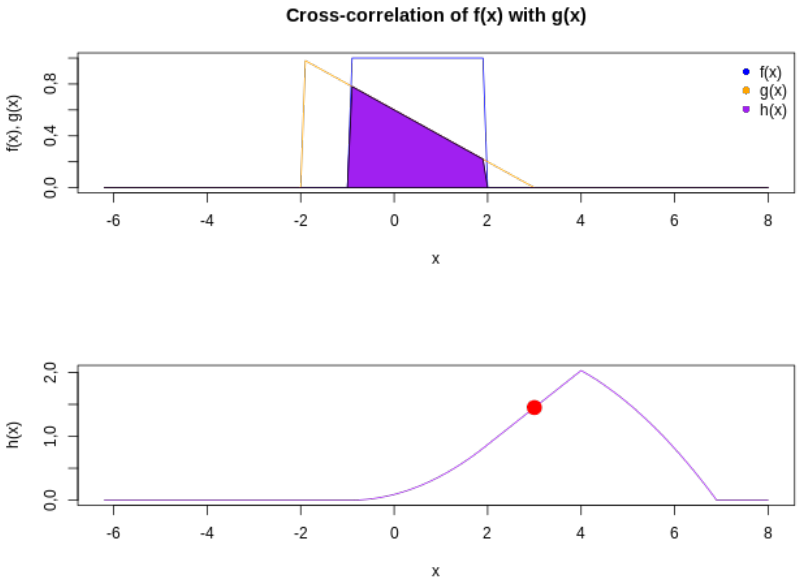


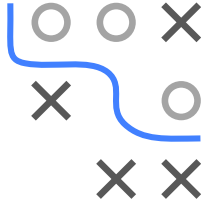
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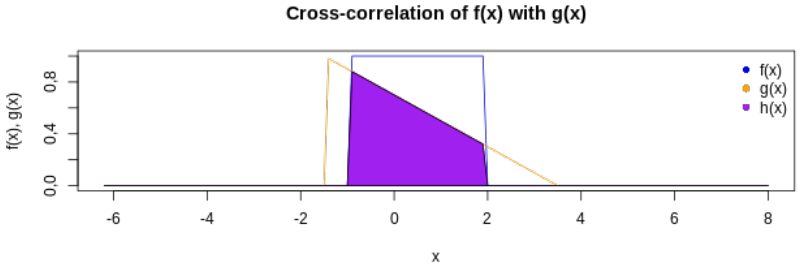


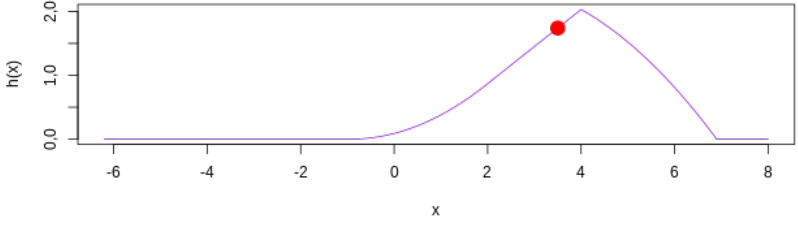
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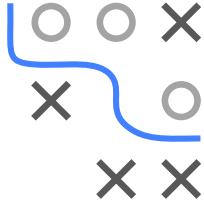




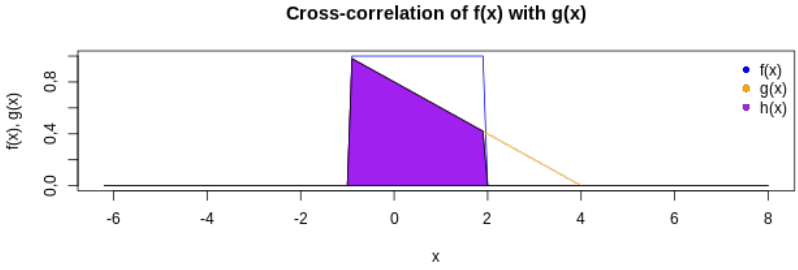
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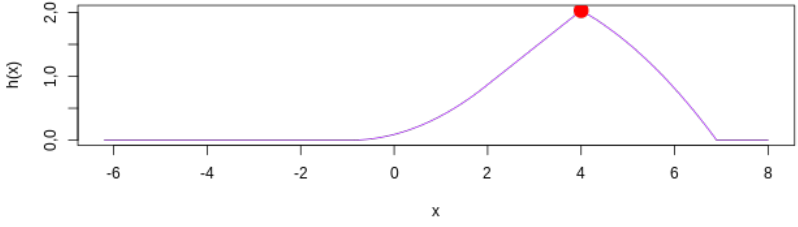


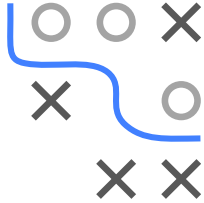




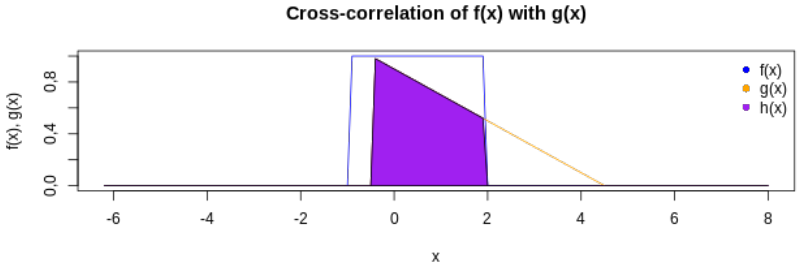
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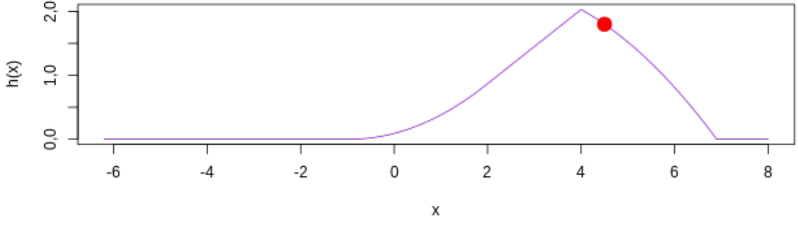


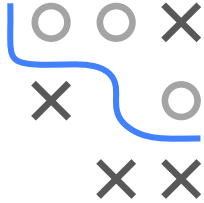




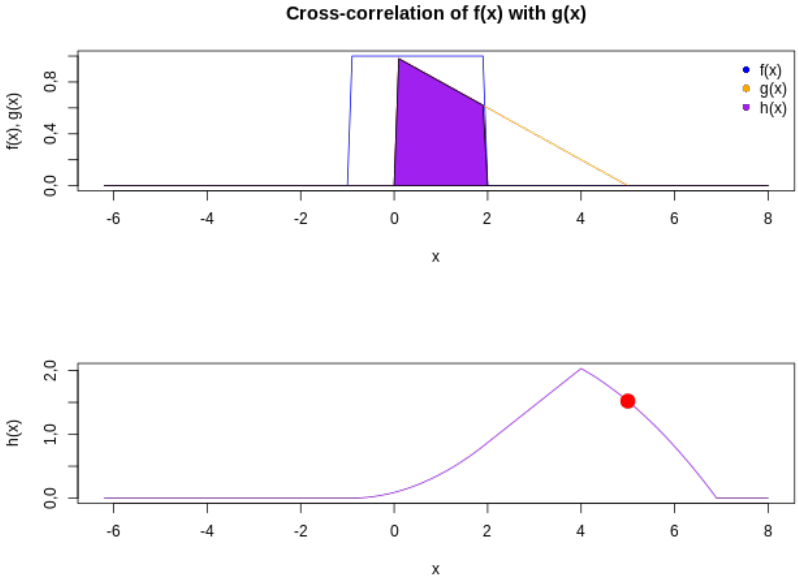
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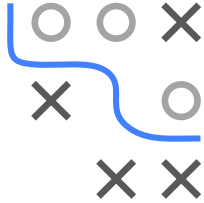




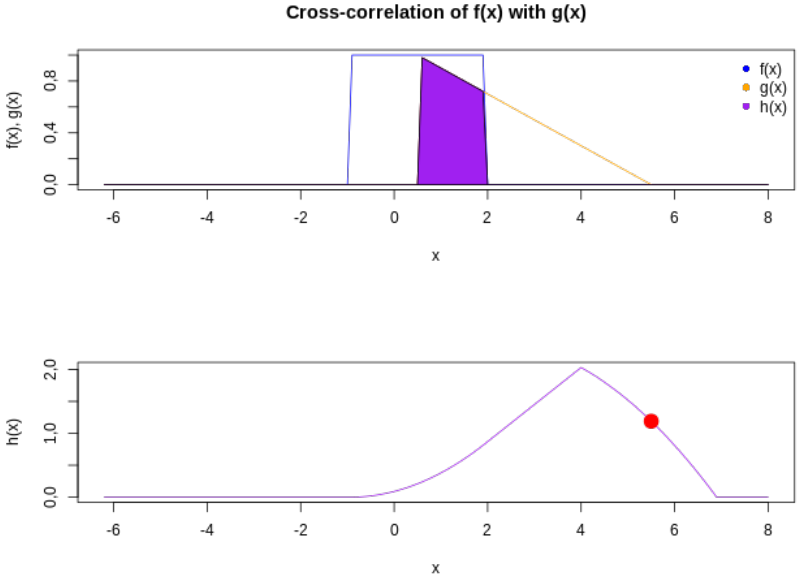


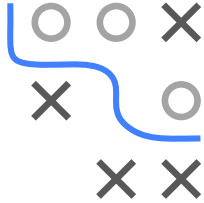
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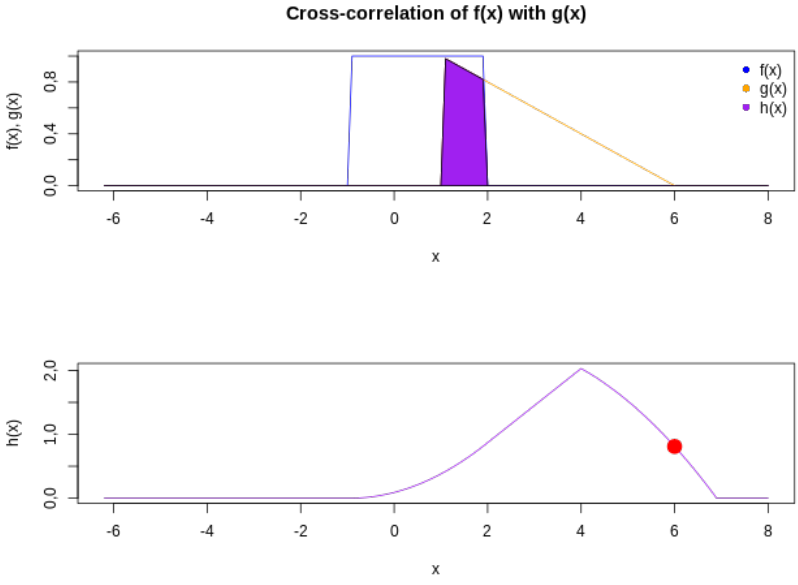


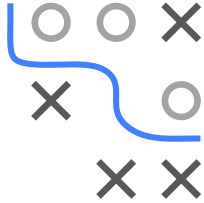
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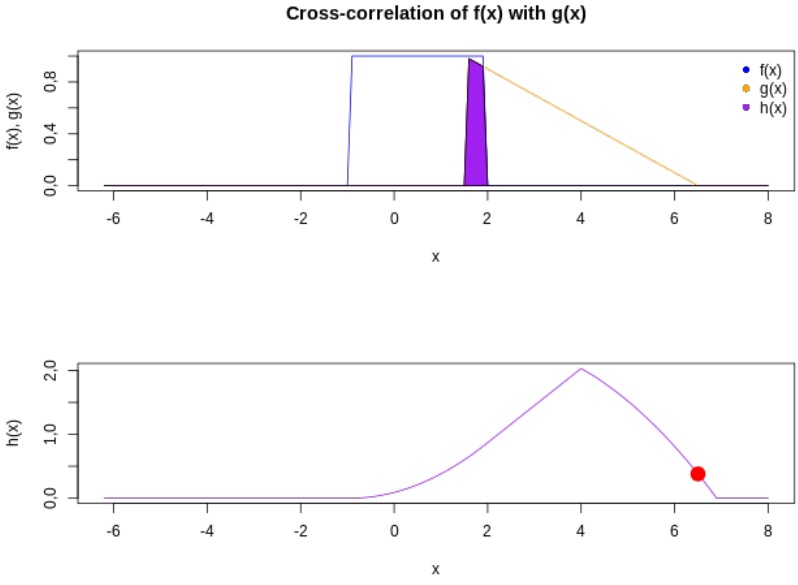


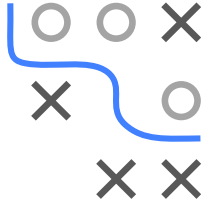
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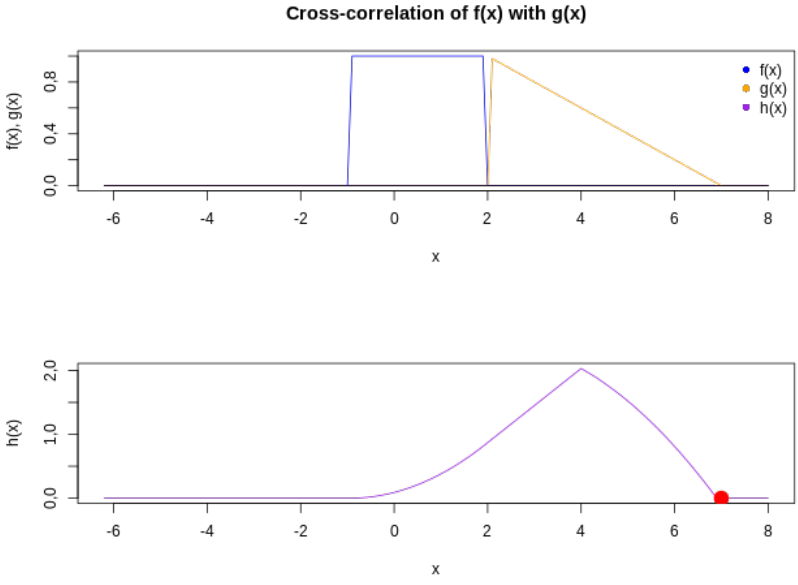


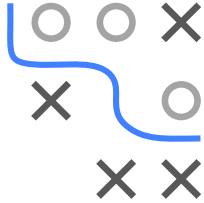
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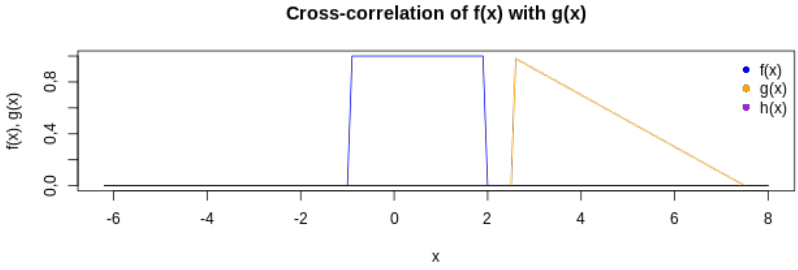


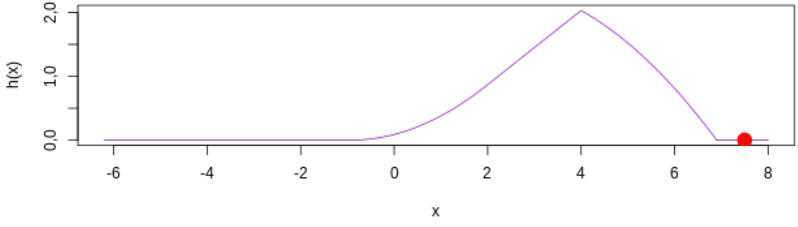
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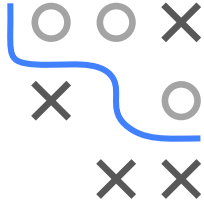




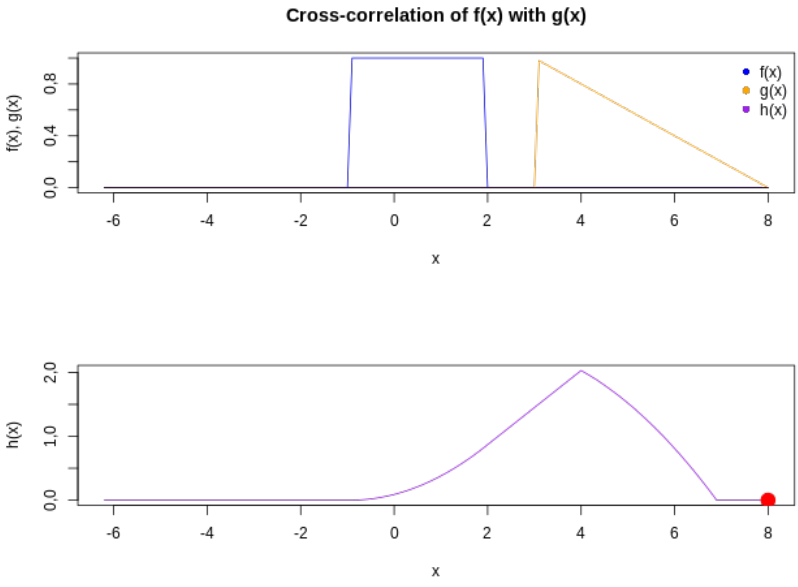
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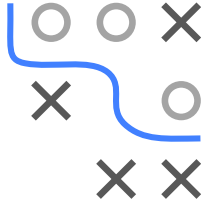






1D CROSS CORRELATION ANIMATION





CROSS CORRELATION VS. CONVOLUTION I

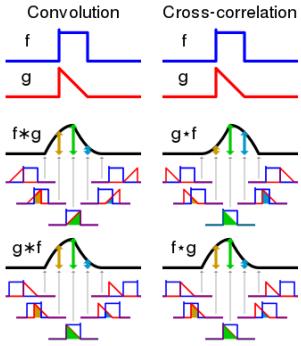
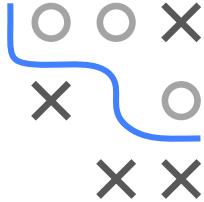


Figure: Comparison of convolution and cross-correlation



- Cross correlation is not commutative
- but often implemented instead of convolution in the practice.